Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

MERN Stack

M1: Practical – 1

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Aim: Environment Setup and JavaScript Basics

a. Install Node.js and Visual Studio Code, and verify installation.

```
PS C:\Users\HAMIZA\Mern> node -v
v10.16.0
PS C:\Users\HAMIZA\Mern> npm -v
6.9.0
```

b. Create a simple Node.js file and run it using the terminal.

Code:

```
prac1.js
console.log("Hello, welcome to 1st practical of mern");
console.log("Afsha")
```

Output:

```
PS C:\Users\HAMIZA\Mern> node prac1.js
Hello, welcome to 1st practical of mern
Afsha
```

c. Write a Node.js script demonstrating basic JavaScript commands (console log, variables, and arithmetic operations).

Code:

```
prac1c.js > ...
console.log("Welcome to node.js");
let num = 80;
let num1 = 40;

console.log("Add=" ,num+num1);
console.log("Sub=" ,num-num1);
console.log("Mul=" ,num*num1);
console.log("Div=" ,num/num1);
console.log("Mod=" ,num%num1);

console.log("Afsha");
```

Output:

```
Welcome to node.js
Add= 120
Sub= 40
Mul= 3200
Div= 2
Mod= 0
Afsha
```

d. Create and run a simple program using variables and functions.

Code:

```javascript
// prac1d.js > ...

let a = 30;
let b = 40;
function Addition(a,b){
    return a + b;

}
console.log("First Number:",a);
console.log("Second Number:",b);
console.log("Add=" ,a+b);
console.log("Afsha");
```

Output:

```
PS C:\Users\HAMIZA\Mern>  node prac1d.js
First Number: 30
Second Number: 40
Add= 70
Afsha
```

e. Demonstrate use of operators and control statements in JavaScript

Code:

```javascript
// prac1e.js > ...
let a = 10;
let b = 20;
// Condition 1
console.log("a > b :", a > b);    // False

// Condition 2
console.log("a < b :", a < b);    // True

// Condition 3
console.log("a == b :", a == b);  // False

// Condition 4
console.log("a != b :", a != b);  // True

// Control Statement
if (a < b) {
    console.log("Condition is True");
} else {
    console.log("Condition is False");
}
console.log("Afsha");
```

Output:

```
PS C:\Users\HAMIZA\Mern>  node prac1e.js
a > b : false
a < b : true
a == b : false
a != b : true
Condition is True
Afsha
```

f. Write programs using loops and template literals.

Code:

```
JS prac1f.js > ...
let num = 4;
console.log(`Multiplication table of ${num}`);
for (let i = 1; i <= 10; i++) {
    console.log(`${num} x ${i} = ${num * i}`);
}
console.log("Afsha");
```

Output:

```
PS C:\Users\HAMIZA\Mern> node prac1f.js
Multiplication table of 4
4 x 1 = 4
4 x 2 = 8
4 x 3 = 12
4 x 4 = 16
4 x 5 = 20
4 x 6 = 24
4 x 7 = 28
4 x 8 = 32
4 x 9 = 36
4 x 10 = 40
Afsha
```